

Byreddy Vasudevareddy

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Summary

Artificial Intelligence and Data Science undergraduate at IIIT Raichur with a strong foundation in software engineering, machine learning, computer vision, and natural language processing. Experienced in building intelligent applications using PyTorch, TensorFlow, LLMs, and modern AI frameworks. Passionate about designing scalable software systems and applying AI to solve real-world challenges.

Relevant Coursework

— Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Probability & Statistics, Linear Algebra.

Projects

XAI Emotion Recognition using Multimodal Feature Fusion and Vision-Language Models 🔗

01/2026 – Present

- Architected a multimodal XAI pipeline fusing ConvNeXt-Tiny vision features with Qwen2.5 language generation to produce grounded, human-readable emotion explanations — a capability absent in standard classifier outputs.
- Encoded FACS Action Units and Grad-ECLIP attention maps as structured VLM prompts, bridging low-level facial geometry with high-level semantic reasoning.
- Achieved **74.06% validation accuracy** on the AffectNet-8 dataset while outperforming ResNet-18 and EfficientNet-B4 baselines.
- Reduced model opacity by grounding predictions in specific facial regions (e.g., brow furrow AU4, lip corner AU12), making outputs auditable for clinical and HCI applications.

Tools Used: Python, PyTorch, ConvNeXt-Tiny, MediaPipe, OpenCV, Hugging Face Transformers, Qwen2.5, Grad-ECLIP

Agentic AI-Powered GitHub Repository Assistant 🔗

08/2025 – 12/2025

- Built an agentic RAG system that answers architecture-level questions about unfamiliar codebases — reducing cold-start comprehension time from hours to minutes.
- Built a semantic code retrieval pipeline using SentenceTransformers and FAISS for efficient repository-aware search.
- Fed NetworkX dependency graphs as structured context into Gemini 2.5 Flash, enabling the agent to explain inter-module dependencies and call chains — not just individual functions.

Tools Used: Python, Gemini 2.5 Flash, SentenceTransformers, FAISS, LangChain, Streamlit, NetworkX

Sign Language to Text Detection System 🔗

08/2024 – 12/2024

- Built a real-time gesture recognition system processing live webcam frames at inference speeds suitable for fluid conversation — no latency lag between gesture and text output.
- Built a gesture-to-text conversion pipeline to improve communication accessibility for hearing-impaired users.

Tools Used: Python, OpenCV, MediaPipe, TensorFlow, NumPy

Positions of Responsibility

Public Relations Team – Shutter Squad Member, <i>Management Team</i> • Captured and documented institute events through photography, supporting official media coverage and digital content creation. • Collaborated with the Public Relations team to produce visual content for institute communications and social media platforms.	07/2024 – 08/2025 IIIT Raichur
Xposure Club, <i>Management Team</i> • Assisted in planning and coordinating technical and cultural events, ensuring smooth execution of club activities. • Worked with the organizing team to manage event logistics, participant coordination, and on-site operations.	09/2023 – 08/2024 IIIT Raichur

Education

Indian Institute of Information Technology Raichur, <i>B.Tech in Artificial Intelligence & Data Science</i> CGPA: 7.0/10	09/2023 – 05/2027 Karnataka, Raichur
Narayana Junior College, <i>Class XII</i>	06/2021 – 04/2023 vijayawada
Viswabharati High School, <i>Class X</i>	04/2021 Gudivada, Krishna

Technical Skills

Programming Python, C, C++, SQL	Software Development: Object-Oriented Programming, Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks
AI & Data Science: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Explainable AI (XAI), Retrieval-Augmented Generation (RAG), Agentic AI.	Frameworks & Libraries: PyTorch, TensorFlow, Keras, Hugging Face Transformers, OpenCV, Scikit-learn, Pandas, NumPy, Matplotlib, MediaPipe, LangChain.
Developer Tools: VS Code, Google Colab, Kaggle.	

Certificates

- NPTEL – Cloud Computing (**Elite + Silver**)
- Infosys Springboard – Introduction to Artificial Intelligence
- Infosys Springboard – Introduction to Data Science

Achievements

- Secured **95.735 Percentile** in JEE Main.
- Scored **97.2%** in Class XII.
- Achieved **10.0 CGPA** in Class X.

Research Interests

- Intelligent Software Systems
- Machine Learning
- Computer Vision
- Natural Language Processing
- Explainable Artificial Intelligence (XAI)